

# Digital Material Passport

ID HT-TEST-001 - Version 0.1.0  
Issue Date 2024-11-15 - Certificate Type EN 10204 3.1

| Manufacturer                                                               | Customer                                                                    | Business Transaction                                                                         |
|----------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Steel Heat Treatment GmbH<br>123 Industrial Park<br>40210 Düsseldorf<br>DE | Automotive Components Inc<br>456 Manufacturing Drive<br>Detroit 48220<br>US | Order ID <b>ORD-2024-HT-047</b><br>Delivery ID <b>DEL-2024-047</b> - Quantity <b>4750 kg</b> |

## Product

|                    |                                |                     |                                     |
|--------------------|--------------------------------|---------------------|-------------------------------------|
| Product Name       | Carbon Steel Round Bar 42CrMo4 | Name (EN 10027-1)   | 42CrMo4                             |
| Batch ID           | BATCH-2024-047                 | Number (EN 10027-2) | 1.7225                              |
| Delivery Condition | +QT - Quenched and Tempered    | Name (AISI/SAE)     | 4140                                |
|                    |                                | Shape               | RoundBar - D 80 mm - Length 6000 mm |

## Heat Treatment

Process: **Quenching and Tempering** - Lot: HT-2024-11-15-B47 - Furnace: FURNACE-03 - Date: 2024-11-15

| Stage         | Temperature | Duration | Cooling | Atmosphere |
|---------------|-------------|----------|---------|------------|
| Austenitizing | 850 C       | 45 min   |         | Air        |
| Quenching     | 60 C        | 10 min   | Oil     |            |
| Tempering     | 550 C       | 120 min  | Air     |            |

## Mechanical Properties

| Property          | Symbol | Actual  | Minimum | Maximum  | Method        | Status |
|-------------------|--------|---------|---------|----------|---------------|--------|
| Tensile Strength  | Rm     | 985 MPa | 900 MPa | 1100 MPa | EN ISO 6892-1 | ✓      |
| Yield Strength    | Re     | 835 MPa | 750 MPa |          | EN ISO 6892-1 | ✓      |
| Elongation        | A      | 14.5 %  | 10 %    |          | EN ISO 6892-1 | ✓      |
| Rockwell Hardness | HRC    | 33 HRC  | 28 HRC  | 35 HRC   | ASTM E18      | ✓      |

## Validation

We hereby certify that the material described above has been heat treated and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated by **Dr. Klaus Schmidt**, Quality Manager,  
Quality Assurance - 2024-11-15